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Dynamics of tax evasion with back auditing, social norm updating, and public goods provision – An agent-based simulation

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ABSTRACT

Income tax evasion dynamics and social interactions are analyzed with an agent-based model in heterogeneous populations. One novelty is the combined analysis of back auditing and ageing, which allows for incorporating psychological findings with respect to social norm updating over a taxpayer's life cycle. Another novelty concerns individual's social behavior regarding a Pareto-optimal provision of public goods. Simulation results support the counterintuitive conclusion drawn elsewhere in the literature that income tax compliance may decrease with raising marginal per capita returns (MPCRs). Yet, back auditing seems to have by far the strongest impact on the dynamics of fiscal fraud and also can help to curb the extent of tax evasion (ETE).

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1. Introduction

Following the lead of Mittone and Patelli (2000), agent-based tax evasion models are gaining more and more popularity. Bloomquist (2006) and Pickhardt and Seibold (2011) provide an overview of recent developments, while Garson (2009) and Heath, Hill, and Ciarallo (2009) offer more general surveys of agent-based models. One reason for the ongoing attractiveness of agent-based tax evasion frameworks may be the fact that these settings give researchers a higher degree of control than available with human subjects in lab experiments. Another distinguishing feature of agent-based modeling is the opportunity to include rather complex real world situations, for instance, various social interaction procedures in a society of behaviorally heterogeneous agents; aspects which the standard neoclassical approach to tax evasion cannot handle.

In this paper heterogeneous taxpayer agents have to file a tax return in an environment which consists of a government, a tax authority and individual social networks of acquaintances. Income tax evasion dynamics are then investigated according to five aspects, that are, (i) lapse of time effects (or back auditing), (ii) age heterogeneity of agents, (iii) social norm evolution

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due to age effects, (iv) provision of pure public goods, and (v) behavioral updating with respect to Pareto-optimal provision of pure public goods. Among other things, the agent-based framework permits to examine various aspects of conditional cooperation and social behavior patterns that may be linked to contributions of many other scholars, e.g. Traxler (2010), Kube and Traxler (2011), Traxler and Spichtig (2011), and Traxler and Winter (2012). Findings include support for the counterintuitive result of Falkinger (1988, 1995) and Cowell (1992) that increasing the marginal per capita return (MPCR) may lead to more tax evasion.

Moreover, it is worth emphasizing that even though agent-based tax compliance models allow for a large variety of simulation scenarios, one may observe at least two open issues. First, the relevance of Pareto-optimal allocations has not yet been analyzed with an agent-based computational setting taking into account tax evasion and public goods provision.[1] Second, there is a persistent lack of docking and replication studies that verify and validate simulation results in this field of research.[2] Therefore, the purpose of this paper is to analyze income tax evasion dynamics within an agent-based model, which addresses both issues by extending and replicating an existing agent-based tax compliance framework.

The paper is organized as follows. The next section presents a mathematical background to augment the neoclassical model of Allingham and Sandmo (1972) by pure public goods provision. The third section describes the agent-based computational model. The fourth section provides results of various simulation scenarios. The final section concludes.

2. Neoclassical foundation

During the last four decades tax evasion models have been developed that add pecuniary as well as non-pecuniary parameters to well-known tax evasion parameters (audit probability, income, tax and penalty rate), and use a wide range of alternative utility functions. Alm, Cronshaw, and McKee (1993), Engel and Hines (1999), Balsa et al. (2006), Antunes, Balsa, Respício, et al. (2007), Hokamp and Pickhardt (2010), and, recently, Seibold and Pickhardt (2013) have contributed to the comparatively little literature about tax evasion and lapse of time effects (related to audits of more than one tax relevant period). For instance, exponential utility functions do allow for back auditing and, therefore, this section focuses on how to incorporate public goods provision into this class of functions.

Beck, Davis, and Jung (1991) was the first study to connect the neoclassical expected utility approach of Allingham and Sandmo (1972) with an exponential utility function in a concept of tax declaration under income uncertainty. Marchese and Privileggi (1999) and Hokamp and Pickhardt (2010) make use of exponential utility functions in an expected utility income tax amnesty setting and an agent-based income tax evasion model, respectively. Therefore, the latter framework is chosen in this paper to create not only a tax compliance docking study using another programming language, but to develop also a computational platform that permits to investigate additional features, *ceteris paribus* and jointly; for instance, social norm updating, public goods provision and lapse of time effects.

The next subsection introduces the provision of pure public goods as an extension of the chosen basic framework. Thereafter, comparative statics and lapse of time effects are considered.

2.1. Exponential utility and public goods provision

Following Hokamp and Pickhardt (2010, Eq. (1), p. 544) an artificial simulation world is populated with a society of N heterogeneous agents with an exponential utility function defined as,

$$U_i[ATPNI_i] = 1 - e^{-\lambda_i \cdot ATPNI_i}. \tag{1}$$

In Eq. (1) λ_i denotes a constant absolute risk parameter and $ATPNI_i$ is the received after tax and penalty net income.[3] The latter variable reflects the outcome of the tax regime and depends on individual taxes, T_i, individual penalties, P_i, and, a public good, X, that is provided according to,

$$X = T + P - A. \tag{2}$$

With respect to (2) the public goods provision, X, is funded by overall tax revenues, T, and penalties, P (which include fines and reimbursement of evaded tax), minus tax administration costs, A. Functional relations concerning the latter three variables due to an individual's tax, T_i, and penalty, P_i, are given by,

$$T = \sum_{i=1}^{N} T_i, \tag{3}$$

[1] Five agent-based tax evasion frameworks consider public goods provision, that are, (i) Mittone and Patelli (2000), (ii) Antunes, Balsa, Urbano, Moniz, and Roseta-Palma (2006), Antunes, Balsa, and Coelho (2007), Antunes, Balsa, Respício, and Coelho (2007), Balsa, Antunes, Respício, and Coelho (2006), (iii) Szabó, Gulyás, and Tóth (2008), Szabó, Gulyás, and Tóth (2009), Szabó, Gulyás, and Tóth (2010), (iv) Pellizzari and Rizzi (2011), and (v) Méder, Simonovits, and Vincze (2012).

[2] Lima (2010), Lima (2012), Andrei, Comer, and Koehler (2011), and Pickhardt and Seibold (2011) are early contributions to agent-based tax evasion docking and replication studies.

[3] The right-hand-side of (1) may take any real value up to 1. For instance, negative utility may occur due to lapse of time effects when accumulated penalties and taxes are greater than actual income and public goods provision. However, how subjects finance a negative outcome ($ATPNI_i$) is not the purpose of this paper. Finally, note that in (1) throughout an agent's life cycle the risk preference (i.e. λ_i) may change, e.g. due to (social) norm evolution (e.g. see Nordblom & Žamac, 2012, and Traxler & Winter, 2012). For further details see Sections 3 and 4.

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$$P = \sum_{i=1}^{N} P_i, \tag{4}$$

and

$$A = \varepsilon_1 \cdot T + \varepsilon_2 \cdot P + \varepsilon_3, \tag{5}$$

where (5) makes use of three administrative function parameters, that are ε_1, ε_2, and ε_3.In particular, $\varepsilon_1 \geqslant 0$ and $\varepsilon_2 \geqslant 0$ captures the effectiveness of collecting taxes and auditing taxpayers, respectively, whereas $\varepsilon_3 \geqslant 0$ represents fixed costs due to tax administration (e.g. see Slemrod & Yitzhaki 2002).[4] Further, for any agent-based simulation run $\varepsilon_1 \leqslant \varepsilon_2 \leqslant 1$ is implicitly assumed due to the fact that collecting taxes is less cost-intensive than enforcing and auditing taxpayers.[5]

Falkinger (1988, 1995) and Cowell (1992) model the notion of taxpayers that their tax evasion behavior is negligible with respect to the prevailing provision level of public goods. Hence, X_{-i} indicates public goods provision without any influence of the corresponding taxpayer agent. To put this differently, X_{-i} neglects impacts on the public goods provision level due to i-th individual's taxes, T_i, and penalties, P_i. Thus, the system (2)–(5) yields,

$$X_{-i} = (1-\varepsilon_1) \cdot \sum_{j=1; j\neq i}^{N} T_j + (1-\varepsilon_2) \cdot \sum_{j=1; j\neq i}^{N} P_j - \varepsilon_3 = \begin{cases} X - (1-\varepsilon_1) \cdot T_i, & \text{if not audited} \\ X - (1-\varepsilon_1) \cdot T_i - (1-\varepsilon_2) \cdot P_i, & \text{if audited} \end{cases} \tag{6}$$

Taxes and penalties depend, among other things, on the decision variable, x_i, which is voluntarily declared income, and the outcome of an auditing procedure. For simplicity, audits are modeled as follows: In any period the tax authority audits each taxpayer at random with probability p. During an audit the tax authority learns an individual's true income (or wage), W_i, and a penalty on income understatement is due.[6] Taxpayers then pay taxes according to,

$$T_i = \theta \cdot x_i, \tag{7}$$

and, in addition, subjects are left with two feasible consequences of a random audit, that are,

$$P_i = \begin{cases} 0, & \text{if not audited} \\ \pi \cdot (W_i - x_i), & \text{if audited} \end{cases} \tag{8}$$

where θ and π are the tax and penalty rate, respectively, with $\theta < \pi$. Finally, taking into account public goods provision the after tax and penalty net income amounts to,

$$ATPNI_i = \begin{cases} Y_i, & \text{if not audited} \\ Z_i, & \text{if audited} \end{cases} \tag{9}$$

where

$$\begin{aligned} Y_i &= W_i - \theta \cdot x_i + \beta \cdot (\theta \cdot (1-\varepsilon_1) \cdot x_i + X_{-i}), \\ Z_i &= W_i - \theta \cdot x_i - \pi \cdot (W_i - x_i) + \beta \cdot (\theta \cdot (1-\varepsilon_1) \cdot x_i + \pi \cdot (1-\varepsilon_2) \cdot (W_i - x_i) + X_{-i}), \end{aligned} \tag{10}$$

corresponds to Allingham and Sandmo (1972, Eq. (2), p. 325). In Eq. (10) $\beta \geqslant 0$ represents the effectiveness of the public sector.[7] Thus, the model incorporates the provision of pure public goods that are non-rival and non-excludable in consumption. Further, for notational convenience two marginal per capita returns (MPCRs) are used that may reflect voluntary and forced contributions to pure public goods according to,

[4] Alternatively, one may assume tax administrative costs, that are $A = \varepsilon_1 N + \varepsilon_2 \cdot p \cdot N + \varepsilon_3$, where N is the number of taxpayers, p is the audit probability and the coefficients ε_1, ε_2, and ε_3 are defined as in (5). Again, costs of processing voluntary tax payments are captured by the first term on the right-hand-side, the second term reflects costs of auditing taxpayers, and the third term, ε_3, are fixed costs. The author thanks an unknown referee for suggesting this modeling. But agent-based simulations frequently run with fixed numbers of taxpayers to allow for *ceteris paribus* conditions, so that the population size N would have no substantial impact on tax evasion dynamics (notable exceptions are Bloomquist, 2004, and Szabó et al., 2010). The present framework (as specified by the system (2)–(5)) permits to examine individual's (rational) reaction on tax cheating and compliance behavior within the context of public goods (e.g. see Falkinger, 1988, for an early theoretical analysis).

[5] For instance, one may think about the effects of raising a flat income tax. Then, *ceteris paribus* tax administration costs do rise because of higher tax revenues due to an increase of (voluntary) paid taxes (collecting taxes is not without costs; e.g. see Slemrod & Yitzhaki, 2002). Higher tax rates may imply stronger incentives to evade taxes (for instance, suppose negative aggregate income and substitution effects according to Allingham & Sandmo, 1972); such an outcome may lead to higher tax administration costs because of substantial monetary changes due to tax authority's audit and deterrence efforts to combat upcoming tax non-compliance. However, magnitudes depend on additional costs that are perhaps not linear. Further, circumstances of institutional changes can matter for affecting tax evasion by law (e.g. see Kube & Traxler, 2011). Helpful comments of two anonymous referees are gratefully acknowledged.

[6] Davis, Hecht, and Perkins (2003), Bloomquist (2004, 2008, 2011), Korobow, Johnson, and Axtell (2007), Lima and Zaklan (2008), Zaklan, Lima, and Westerhoff (2008), Zaklan, Westerhoff, and Stauffer (2009), Szabó et al. (2008, 2009, 2010), Lima (2010, 2012), Hokamp and Pickhardt (2010), Andrei et al. (2011), Pellizzari and Rizzi (2011), Pickhardt and Seibold (2011), and, recently, Seibold and Pickhardt (2013) do implement random audit schemes. Further, Mittone and Patelli (2000) and Bloomquist (2011) consider alternative inspection settings such as 'lower tail' and 'discriminant function (DIF) style' auditing, respectively. Moreover, Bloomquist (2004, 2006, 2008), Antunes et al. (2006), Antunes, Balsa, and Coelho (2007), Antunes, Balsa, Respício, et al. (2007), and Balsa et al. (2006) explicitly examine the role and the behavior of tax inspectors. Econophysics tax evasion models frequently apply a penalty for tax offenses that consists of a pre-defined full compliance period whereas economics frameworks often deal with monetary penalties.

[7] $\beta = 0$ allows to examine frameworks without any kind of public goods provision (e.g. Hokamp & Pickhardt, 2010). $\beta < 0$ implements a public bad; such a setting is ruled out throughout this paper.

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$$MPCR_{voluntary} = \beta \cdot (1 - \varepsilon_1), \tag{11}$$

and,

$$MPCR_{forced} = \beta \cdot (1 - \varepsilon_2), \tag{12}$$

respectively. Then, $\Delta_i[x_i]$ describes a change of the after tax and penalty net income due to detected and uncovered tax non-compliance, that is,

$$\Delta_i[x_i] = Y_i - Z_i = (1 - MPCR_{forced}) \cdot \pi \cdot (W_i - x_i). \tag{13}$$

In particular, to guarantee that uncovered tax non-compliance does not pay, the term on the right-hand-side (henceforth r.h.s.) of (13) has to be strictly positive. Hence, $1 - MPCR_{forced} > 0$ and $x_i < W_i$ fulfill the latter condition and, therefore, detected tax evasion has no (positive) value. However, other feasible options are thinkable to ensure that the r.h.s. of (13) is positive, but preconditions in the manner chosen above do avoid degenerated cases (e.g. tax overpayment, absence of prisoner's dilemma situations and the like). For instance, a prisoner's dilemma situation applies if,

$$\frac{1}{N} < MPCR_{voluntary} < 1 \tag{14}$$

(e.g. see Croson 2007, p. 200, and Hokamp & Pickhardt 2011, Eq. (6), p. 3).[8] Given the constraints above and then applying neoclassical expected utility theory yields an interval for the audit probability p representing a necessary and sufficient condition for an inner solution,[9]

$$\frac{(1 - MPCR_{voluntary}) \cdot \theta}{(1 - MPCR_{voluntary}) \cdot \theta + ((1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta) \cdot e^{\lambda_i \cdot \Delta_i[0]}} < p < \frac{(1 - MPCR_{voluntary}) \cdot \theta}{(1 - MPCR_{forced}) \cdot \pi}. \tag{15}$$

Therefore, the inner solution is calculated with respect to,

$$x_i = W_i - \frac{\ln[\{(1-p) \cdot (1 - MPCR_{voluntary}) \cdot \theta\} / \{p \cdot ((1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta)\}]}{(1 - MPCR_{forced}) \cdot \lambda_i \cdot \pi}, \tag{16}$$

which extends Hokamp and Pickhardt (2010, Eqs. (5) and (6), p. 545) in line with Allingham and Sandmo (1972, Eqs. (5) and (6), p. 325). Again, following Hokamp and Pickhardt (2010, p. 545), according to (15), taxpayer agents are fully compliant, that is $x_i = W_i$, if p exceeds the upper bound, and subjects do totally conceal their tax liabilities, that is $x_i = 0$, if p falls below the lower bound. However, the next subsection presents comparative statics to inspect the dependence of the inner solution, x_i, on model parameters.

2.2. Comparative statics

For the sake of clarity, see the mathematical appendix for partial derivatives missing below. Regarding tax evasion parameters income (or wage) W_i, tax rate θ, penalty rate π, and audit probability p the results are confirmed as predicted by Allingham and Sandmo (1972). To put it in other terms, voluntarily declared income, x_i, rises with increasing individual income W_i, penalty rate π, and audit probability p. According to Allingham and Sandmo (1972, Eq. (14), p. 329) the income effect is zero due to constant absolute risk aversion (henceforth CARA) of underlying exponential utility functions. In addition, the substitution effect is negative, so that voluntarily declared income x_i decreases while increasing the tax rate θ.

Moreover, the framework may also serve as an example with respect to the extension of Yitzhaki (1974); fines only depend on evaded tax, i.e. $\pi = F \cdot \theta$, where $F > 1$ is a factor of punishment. As predicted by Yitzhaki (1974, Eq. (14*), p. 202), the opposite result occurs, i.e. x_i grows while increasing θ; an outcome that reflects a positive pure income effect. No change is observable regarding the fine, so that x_i increases when raising F. Finally, x_i rises when increasing the risk parameter λ_i due to CARA.

Furthermore, this study of income tax evasion dynamics differs from other agent-based tax compliance and evasion models, among other things, with respect to the remaining parameters, that are, the effectiveness of the public sector β, and the three administrative function parameters, ε_1, ε_2, and ε_3. To examine an Allingham and Sandmo (1972) framework with pure public goods provision, that is, $\beta > 0$, comparative statics below allow for new insights. To begin with, x_i decreases with increasing the tax collection costs, that is, the effectiveness parameter ε_1.This conclusion is based on,[10]

$$\frac{\partial x_i}{\partial \varepsilon_1} = -\frac{\beta}{(1 - MPCR_{forced}) \cdot ((1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta) \cdot \lambda_i} < 0. \tag{17}$$

[8] Croson (2007) considers (14) within homogeneous parameter settings. Following Hokamp and Pickhardt (2011) this constraint degenerates and represents a necessary condition if heterogeneous endowments are considered. Then, a sufficient condition for a prisoner's dilemma to arise is needed, that is, $W_i/W < MPCR_{voluntary}$, depending on individual's income W_i, total endowment of the population, $W = \sum_{i=1}^{N} W_i$, and marginal per capita return due to voluntary contributions, $MPCR_{voluntary}$. In addition, note that, $MPCR_{forced} \leqslant MPCR_{voluntary} < 1$, applies because of $\varepsilon_1 \leqslant \varepsilon_2$. However, throughout the paper necessary and sufficient conditions for prisoner's dilemma situations are fulfilled in any simulation scenario.

[9] Recognize that $(1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta > 0$ allows for an inner solution since (15) is violated otherwise.

[10] Recall,$1 - MPCR_{forced} > 0$ and $(1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta > 0$.

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An opposite result occurs for the effectiveness parameter ε_2, that is, x_i increases while raising the auditing costs. This statement is due to,

$$\frac{\partial x_i}{\partial \varepsilon_2} = \frac{\beta}{(1 - MPCR_{forced})^2 \cdot \lambda_i} \cdot \left(\frac{1 - MPCR_{forced}}{(1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta} + \frac{NLT}{\pi} \right) > 0, \tag{18}$$

where *NLT* represents the natural logarithmic term on the r.h.s. of (16).[11] Thus, a rational taxpayer *ceteris paribus* declares less (more) income if tax collection (auditing) costs increase.[12] Further, x_i is obviously independent from fixed tax administration costs ε_3.With respect to the effectiveness of the public sector β (and its influence on x_i) there is in general no clear-cut conclusion available due to,

$$\frac{\partial x_i}{\partial \beta} = \frac{1}{(1 - MPCR_{forced})^2 \cdot \lambda_i} \cdot \left(\frac{(\varepsilon_2 - \varepsilon_1) \cdot (1 - MPCR_{forced})}{(1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta} - \frac{(1 - \varepsilon_2) \cdot NLT}{\pi} \right) \gtreqless 0. \tag{19}$$

To illustrate the notion behind (19) consider two examples: First, the agent-based tax evasion setting of Méder et al. (2012) neglects any influence of penalties, in particular, with respect to pure public goods. Such a framework may be reflected by $\varepsilon_1 = 0$, $\varepsilon_2 = 1$, and $\varepsilon_3 = 0$.[13] Then, x_i increases with raising the effectiveness of the public sector β, according to (19). Second, the experimental tax evasion model of Bazart and Pickhardt (2011) incorporates penalties and a tax lottery payoff L, i.e. $\varepsilon_1 = 0$, $\varepsilon_2 = 0$, and $\varepsilon_3 = L$.[14] An opposite effect turns now out, that is, x_i decreases with increasing the effectiveness of the public sector β. To continue, the next subsection adds back auditing to previous thoughts and analyzes comparative statics within these settings.

2.3. Back auditing

In this section the framework models multiple time periods by extending (9) and (10), that is,

$$ATPNI_{i,t} = \begin{cases} Y_{i,t} = Y_i, & \text{if not audited} \\ Z_{i,t} = Z_i - (1 - MPCR_{forced}) \cdot \sum_{k=t-\alpha}^{t-1} \pi_k \cdot (W_i - x_{i,k}), & \text{if audited} \end{cases} \tag{20}$$

In Eq. (20), α represents a back auditing parameter, that is, the number of previous tax relevant periods t subject to an audit in addition to the actual time period (e.g. see Antunes, Balsa, Respício, et al., 2007, p. 87). Hence, the implementation of pure public goods provision is one novelty, e.g. in contrast to the second scenario of Hokamp and Pickhardt (2010, p. 550). Again, their neoclassical expected utility approach leads to a necessary and sufficient condition for an inner solution, that is,

$$\frac{(1 - MPCR_{voluntary}) \cdot \theta}{(1 - MPCR_{voluntary}) \cdot \theta + ((1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta) \cdot e^{\lambda_i \cdot \Delta_i [-\sum_{k=t-\alpha}^{t-1} \frac{\pi_k}{\pi} (W_i - x_{i,k})]}} < p$$

and, (21)

$$p < \frac{(1 - MPCR_{voluntary}) \cdot \theta}{(1 - MPCR_{voluntary}) \cdot \theta + ((1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta) \cdot e^{\lambda_i \cdot \Delta_i [W_i - \sum_{k=t-\alpha}^{t-1} \frac{\pi_k}{\pi} (W_i - x_{i,k})]}}$$

Thus, the inner solution is given by,

$$x_{i,t} = W_i + \sum_{k=t-\alpha}^{t-1} \frac{\pi_k}{\pi} \cdot (W_i - x_{i,k}) - \frac{\ln[\{(1 - p) \cdot (1 - MPCR_{voluntary}) \cdot \theta\} / \{p \cdot ((1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta)\}]}{(1 - MPCR_{forced}) \cdot \lambda_i \cdot \pi}. \tag{22}$$

It is worth mentioning that (21) and (22) extend (15) and (16), and Hokamp and Pickhardt (2010, Eqs. (10) and (11), p. 550) concerning lapse of time effects, and pure public goods provision, respectively.[15]

[11] According to the necessary and sufficient constraint for an inner solution (15) the natural logarithmic term is non-negative, which is $NLT = \ln[\{(1 - p) \cdot (1 - MPCR_{voluntary}) \cdot \theta\} / \{p \cdot ((1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta)\}] \geqslant 0$. Therefore, tax overpayments are ruled out in the inner solution that is (16), of the tax evasion gamble.

[12] For instance, reconsider (17) where rational expected utility maximizing taxpayers evade more taxes if collection costs do increase. Such a behavior may reflect the notion that decreasing the marginal per capita return, $MPCR_{voluntary}$, *via* increasing ε_1 leads to less tax compliance (see Zelmer, 2003, pp. 304–305).

[13] $X = T + P - A = T + P - P = T$ shows no influence of penalties regarding the prevailing provision level, X, of public goods (see (2)–(5)). To clarify the original setting of Méder et al. (2012) implies $\pi = 0$, a condition that the present model rules out for various reasons (e.g. violating constraints as in expressions (7) and (12), division by zero, and the like). However, without any kind of penalties rational expected utility maximizing taxpayers should declare nothing at all. Comments of an anonymous referee are gratefully acknowledged.

[14] $X = T + P - A = T + P - L$ corresponds to Bazart and Pickhardt (2011, Eq. (3), p. 130). Lottery payoffs are (tax administrative) costs that have to be deducted from financing public goods. The author thanks an anonymous referee.

[15] Again, because of the necessary and sufficient condition for an inner solution (21) the natural logarithmic term is non-negative, which is $NLT = \ln[\{(1 - p) \cdot (1 - MPCR_{voluntary}) \cdot \theta\} / \{p \cdot ((1 - MPCR_{forced}) \cdot \pi - (1 - MPCR_{voluntary}) \cdot \theta)\}] \geqslant 0$. Furthermore, equation (21) guarantees that $0 \leqslant x_{i,t} \leqslant W_i$, which excludes tax overpayments according to (22).

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Comparative statics of the preceding subsection are now revised to incorporate lapse of time effects. Obviously, previous results maintain, except regarding three variables that are, (i) the penalty rate π, (ii) the tax rate θ, and (iii) the fine F; where the latter two settings refer to Yitzhaki (1974). In these three cases the lapse of time effect is negative and leads to no definite conclusion whether at period t voluntarily declared income, $x_{i,t}$, is increasing, constant or decreasing. Furthermore, assuming the absence of public goods provision, that is, $\beta = 0$, comparative statics of sections 2.2 and 2.3 may also apply to the underlying agent-based model of Hokamp and Pickhardt (2010).

To sum up, in many cases no clear-cut conclusion is yet available as shown by comparative statics. Thus, computational simulations can be useful to examine, for instance, effects of pure public goods provision in a context of income tax evasion and social interaction (see Section 4). The next section introduces the basic model and some additional features with a special focus on social interaction dynamics not only among agents themselves, but also with their institutional environment (e.g. see Kube & Traxler, 2011).

3. Model and social interaction dynamics

Within the model three types of agents interact that represent a population of taxpayers, a tax authority, and a government. The latter agent provides pure public goods using total net revenue, which is defined as the sum of taxes and penalties minus tax administration costs according to the system (2)–(5). The tax authority collects taxes and penalties after conducting a random audit procedure (see section 2.1). In addition, for any (inter-) acting taxpayer agent the tax authority announces actual values of model parameters that may reflect a political cycle. Taxpayer agents derive their after tax and penalty net income, which may include an additional payoff due to the provision of pure public goods. Resulting income tax evasion dynamics are then analyzed and compared according to the extent of tax evasion, ETE_t; a key figure that is calculated with respect to the sum of actual voluntarily declared income and overall income,

$$ETE_t = 1 - \frac{\sum_{i=1}^{N} x_{i,t}}{\sum_{i=1}^{N} W_{i,t}}. \tag{23}$$

In particular, following Hokamp and Pickhardt (2010), assume four taxpayer agent subtypes, which are, (i) neoclassical A-type agents, (ii) social interacting B-type agents, (iii) ethical C-type agents, and (iv) erratic D-type agents. The A-types file a tax return after calculating their voluntarily declared income with respect to a neoclassical expected utility maximizing procedure due to Allingham and Sandmo (1972). The rational calculation process of these agents was presented in section 2.1. Further, suppose that A-types misperceive the real audit probability p in some periods after an audit has taken place (e.g. because of fear, guilt, shame, and the like, see Kirchler, 2007). In this case an agent adjusts his/her subjective audit probability to $p_s = 1$ (first period after an audit), and thereafter applies an updating procedure by adding $\delta = -0.20$ until the real audit probability p is reached again, that is, $p = p_s$. The B-types are conditional social interacting taxpayers; first of all, they check if the observed tax evasion behavior of their social network, S (of size v), was successful one period earlier (that is, the tax relevant period $t-1$), which was the case if the following constraint holds,

$$\frac{1}{v} \sum_{s \in S \subseteq \{1,\ldots,N\};|S|=v} \left(1 - \frac{ATPNI_{s,t-1}}{W_{s,t-1}}\right) < \theta_{t-1}, \tag{24}$$

where θ_{t-1} is the relevant tax rate at period $t-1$. Then, provided social network's tax evasion has paid the B-types conceal their tax liabilities according to the average cheating in the past, that is,

$$x_{i,t} = \frac{1}{v} \sum_{s \in S \subseteq \{1,\ldots,N\};|S|=v} x_{s,t-1} \cdot \frac{W_{i,t}}{W_{s,t-1}}. \tag{25}$$

In any other case (e.g. within four tax relevant periods after uncovered tax cheating behavior, which yields subject's mutation period $\tau = 4$) the B-types make use of their default option, i.e., they declare their income truly (see Traxler (2010), Traxler & Spichtig (2011), and Traxler & Winter (2012), regarding social norms, enforcement, and cooperation). With respect to the simulations let the social network be a fixed list of four taxpayers or acquaintances, i.e. $v = 4$ (e.g. for similar frameworks see Davis et al. (2003), and Bloomquist (2004)). The C-types always disclose their true income due to ethical reasons, altruism, or social norms. Finally, the D-types intend to be wholly tax-compliant but they may make errors when filing their tax return because of tax law complexities γ (e.g. see Kube & Traxler (2011) considering 'mild' laws). The voluntarily declared income of a D-type agent is assumed to be normally distributed with expectation $\mu_i = W_i$ and standard deviation $\sigma_i = \gamma W_i$ (for further details see Hokamp & Pickhardt (2010), and Andrei et al. (2011)).[16]

Four extensions are novelties regarding the underlying agent-based setting of fiscal fraud, which are, (i) 'age heterogeneity', (ii) 'social norm updating', (iii) 'public goods provision' and (iv) 'Pareto-optimality'. Each novelty can be switched on or off to allow for examining tax evasion dynamics *ceteris paribus* and jointly.

[16] Following Hokamp and Pickhardt (2010) voluntary declared income of D-types is restricted to values between zero and W_i/θ to avoid degenerated cases and to allow for *ceteris paribus* conditions. Alternatively, Andrei et al. (2011) employ a uniform distributed income where such cases cannot appear.

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First, if 'age heterogeneity' is activated, taxpayer agents are born in the initial period at different ages; at the beginning of any tax relevant period they get one year older and die at a fixed maximum age.[17] To ensure *ceteris paribus* conditions taxpayer agents are replaced one period after their death by agents with identical characteristics, except for age, that is, updated to a minimum age. Note that all agents are born without any kind of history. Further, the number of dying agents in each period is pre-selected at random and not assigned to a fixed number (see Bloomquist, 2011).

Second, when 'social norm updating' is enabled, risk parameters λ_i of the A-types rise in a stepwise fashion while agents grew old; a feature that models an increased risk aversion of the elderly. This setting may reflect an evolution of social norms during an agent's life span (e.g. see Nordblom & Žamac, 2012). In addition, one may argue that the elderly are more conditional cooperative taxpayers (see Traxler, 2010, with respect to social norms, stigmatization, and conditional interaction).

Third, when 'public goods provision' is switched on, the A-types show an expected utility maximizing behavior (as specified in section 2.1) taking the provision of pure public goods into account. The B-types are affected by public goods provision *via* the A-types in their social network. However, C- and D-type decision heuristics are assumed to be not influenced by the provision of pure public goods.

Fourth, if 'Pareto-optimality' is turned on, the B-types search for Pareto-optimality and do not change their voluntarily declared income, provided one tax relevant period earlier the public goods allocation was already Pareto-optimal.[18] Given an absence of Pareto-optimal allocations B-types may change voluntarily declared income according to their underlying decision heuristics and social networks. To phrase it differently, provided the B-types do not find Pareto-optimal allocations they interact with other taxpayer agents through their social network *via* applying a check for successful tax evasion behavior according to (24). Therefore, they may evade taxes or make use of their default option (and declare their taxes fully). However, the next section presents simulation scenarios and results.

4. Simulation scenarios and results

This section analyzes income tax evasion dynamics incorporating lapse of time effects and several changes of tax relevant parameters. To begin with, each simulation scenario makes use of a political cycle of 40 tax relevant periods (see Appendix, Table A1) following Hokamp and Pickhardt (2010, Table 1, p. 544). A heterogeneous population of $N = 2000$ taxpayer agents is endowed at the initial period with heterogeneous incomes which are uniformly distributed between 0 and 100.[19] In addition, initial values of individuals' risk aversion parameter are uniformly distributed between 0 and 1. Alternatively, one may assume that $\lambda_i \in]0; \infty[$; but then conditions are violated to replicate *ceteris paribus* Hokamp and Pickhardt (2010). For the same reason simulation runs are done with 50% A-types, 35% B-types, 0% C-types, and 15% D-types, except runs concerning the last subsection with extension 'Pareto-optimality'. Latter results are obtained with 50% A-types, 35% B-types, 15% C types, and 0% D-types (section 4.2 motivates this choice). On the technical side the paper makes use of Java and the RePast software packages, which allows extending and replicating the MATLAB code of Hokamp and Pickhardt (2010). Java codes are available upon request and, in particular, Random Seed 11223344 permits to reproduce any figure below. Finally, all figures contain two benchmarks that are replications of Hokamp and Pickhardt (2010). In particular, reproductions of 'Scenario 1' and 'Scenario 2' are denoted as 'Replication 1' and 'Replication 2', which run without and with back auditing, i.e. $\alpha = 0$, and $\alpha = 10$, respectively (see Appendix). Table 1 gives an overview simulation features and related figures.

4.1. Age heterogeneity and moral age effects

The first extension, 'age heterogeneity', assumes taxpayer agents to be born at a minimum age of 21 and to die at a maximum age of 60. The second extension, that is, 'social norm updating', increases at random, in a stepwise fashion, the individual's risk parameter λ_i every 10 years. To put it in other terms, λ_i is uniformly distributed depending on an agent's age a_i according to Table 2.

Figs. 1.1 and 1.2 present results of six simulation runs and make use of two replications as benchmarks (see Appendix). To begin with, Fig. 1.1 provides strong evidence that without back auditing social norm updating has an intense influence on the magnitude of tax evasion; although tax evasion dynamics have the same shape in comparison to the benchmark, that is, 'Replication 1' (see 'Simulation 2' and 'Simulation 3'). Further, Figs. 1.1 and 1.2 show that age heterogeneity generates effects on income tax evasion dynamics only under social norm updating or back auditing (inspection of 'Simulation 1' indicates no effects). More precisely, under effects of back auditing age heterogeneity leads to fluctuations with smaller peaks than without the feature (see 'Simulation 4' versus 'Replication 2'). A moral age effect then leads to a substantial vertical downward shift of income tax evasion (see 'Simulation 5' and 'Simulation 6'). However, income tax evasion dynamics driven by param-

[17] By default all agents are born at the initial period, grow one year older at the beginning of any tax relevant period but live forever.

[18] Following Hokamp and Pickhardt (2011) the notion of Pareto-optimality is adopted. Among other things, the authors show that in an artificial world with income heterogeneity Pareto-optimality depends on the maximum number of free-riders, n_{max}, that is connected to the marginal per capita return of voluntary contributions, $MPCR_{voluntary}$ (see cf., p. 26). Note that Hokamp and Pickhardt (2011) make use of the term 'free-rider' in the sense that agents do not fulfill their (tax) obligations totally. To put this differently, free-riders contribute nothing or a smaller amount than required. Therefore, (partial and full) free-riders are the counterpart of full contributors who do not cheat and pay their taxes totally.

[19] This modeling allows for an interesting interpretation. One may identify the lowest and highest income level with 0 and 100, respectively. Of course, income is most likely not uniformly distributed. However, given another income distribution it is possible to adjust and re-run the agent-based tax evasion simulation.

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Table 1
Overview – Simulation specifications.

Feature run	Lapse of time	Age heterogeneity	Social norm updating	Public goods	Pareto-optimality	Figure
Simulation 1	0	On	Off	Off	Off	1
Simulation 2	0	On	On	Off	Off	1
Simulation 3	0	Off	On	Off	Off	1
Simulation 4	10	On	Off	Off	Off	2
Simulation 5	10	On	On	Off	Off	2
Simulation 6	10	Off	On	Off	Off	2
Simulation 7	0	Off	Off	On	Off	3
Simulation 8	0	On	On	On	Off	3
Simulation 9	0	On	On	On	On	3
Simulation 10	10	Off	Off	On	Off	4
Simulation 11	10	On	On	On	Off	4
Simulation 12	10	On	On	On	On	4
Replication 1	0	Off	Off	Off	Off	All
Replication 2	10	Off	Off	Off	Off	All

Note: The specifications of 14 simulation runs, i.e. Simulations 1–12, and Replications 1 and 2, are given with respect to five features, which are, 'lapse of time effects', and the four novelties, 'age heterogeneity', 'social norm updating', 'provision of pure public goods', and a check for 'Pareto-optimality'. 'on' marks active features. 'off' indicates options that are switched off. Without and with lapse of time effects is represented by corresponding tax relevant periods subject to an audit, that are $\alpha = 0$, and $\alpha = 10$. 'Figure' denotes the graphic representation(s) that displays results of the corresponding simulation run. 'all' indicates benchmarks that are presented in any figure, i.e. in Figs. 1.1, 1.2, 2.1 and 2.2.

Table 2
Social norm updating (A-types).

Age a_i	$20 < a_i < 31$	$30 < a_i < 41$	$40 < a_i < 51$	$50 < a_i < 61$
Risk parameter λ_i	[0; 0.25]	[0.25; 0.50]	[0.50; 0.75]	[0.75; 1]

Note: λ_i denotes ith agent's risk parameter, that is uniformly distributed for utility maximizing A-type taxpayers (according to four intervals) depending on the corresponding age a_i (between 21 and 60), if and only if the extension 'social norm updating' is activated.

eter changes over time (e.g. tax and penalty rates may differ due to the underlying political cycle of Hokamp & Pickhardt, 2010) remain remarkable, especially under back auditing. Finally, simulations have shown substantial interaction dynamics due to both extensions, age heterogeneity and social norm updating (see 'Simulation 2' and 'Simulation 5'); and upon other terms the next section now examines these settings.

4.2. Public goods provision and Pareto-optimality

Within in this subsection all simulation runs are done under the extension 'public goods provision'. In particular, neoclassical A-types are influenced by effectiveness of the public sector β, and administrative function parameters ε_1, ε_2, and ε_3. These governmental policy parameters may change over time due to a public goods provision cycle of Table 3.[20]

The B-types consider the (Pareto-optimal) provision of pure public goods *via* their social network. Note that Pareto-optimality under tax overpayment is still an open issue (see Hokamp & Pickhardt, 2011). Therefore, simulations with the extension 'Pareto-optimality' make use of 50% A-types, 35% B-types, 15% C-types, and 0% D-types, to rule out tax overpayment. Since the D-types declare their true income on average and the C-types always declare their true income, Fig. 2.1, 'Simulation 9', and Fig. 2.2, 'Simulation 12', are comparable, but only to a limited extent. However, Figs. 2.1 and 2.2 visualize simulation results regarding pure public goods provision and behavioral updating due to Pareto-optimality, respectively.

Fig. 2.1 shows that neglecting back auditing may lead to situations where raising public goods benefits (MPCR's) increases income tax evasion (see 'Simulation 7' versus 'Replication 1'). Even this counterintuitive finding with respect to the provision of pure public goods is in line with Falkinger (1988, 1995) and Cowell (1992). It is worth noting that, under back auditing, these dynamics are still present but turn to be rather marginal; for instance, see Fig. 2.2, 'Simulation 10' and 'Replication 2', Periods 37–40. However, age heterogeneity and social norm updating still work to curb tax evasion (see 'Simulation 8'). Further, in comparison to the latter simulation run effects due to Pareto-optimality are more likely marginal ('Simulation 8' versus 'Simulation 9'). In contrast, according to Fig. 2.2, fluctuations of income tax evasion dynamics indicate that from period 9–16 Pareto-optimal allocations exist, which lead to behavioral reactions one time step later. In these periods inspection of Table 3, column 'Maximum number of free-riders', yields that the Pareto-optimality concept just tolerates up to about 1000 taxpayer agents, which may deviate from true income declaration. In comparison to age driven norm updating, behavioral responses to Pareto-optimality may increase tax evasion (e.g. see 'Simulation 11' versus 'Simulation 12').

[20] Parameters β, ε_1, ε_2, and ε_3 can be freely selected. For instance, Slemrod and Yitzhaki (2002, p. 1448–1449) provide magnitudes for ε_1. However, remaining values are selected to reflect a political cycle of a government that takes less or more care of public goods provision and tax administration costs. Note that tax administration costs do influence remarkable Pareto-optimality *via* the voluntary marginal per capita return, $MPCR_{voluntary}$.

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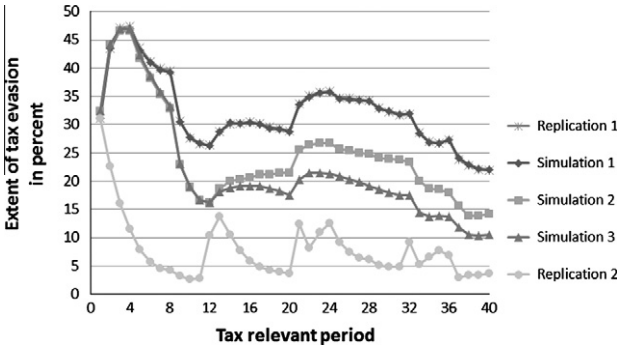


Fig. 1.1. Age heterogeneity and social norm updating ($\alpha = 0$).

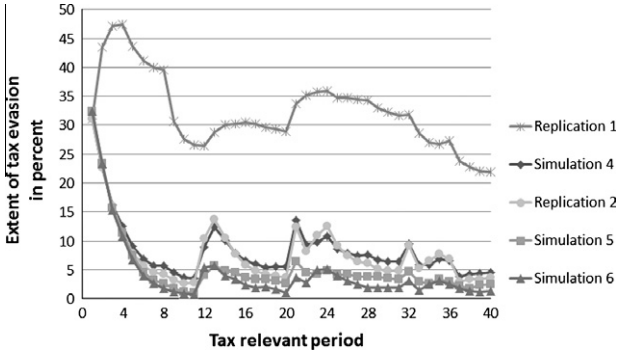


Fig. 1.2. Age heterogeneity and social norm updating ($\alpha = 10$). *Note*: Figure 1.1 shows the extent of tax evasion without back auditing under 'age heterogeneity' (Simulation 1), 'social norm updating' (Simulation 3), and both extensions (Simulation 2). Figure 2 presents the extent of tax evasion with back auditing under 'age heterogeneity' (Simulation 4), 'social norm updating' (Simulation 6), and both extensions (Simulation 5). In addition, two benchmarks are provided, that are, Replications 1 and 2 reflecting reproductions of Scenarios 1 (without back auditing) and 2 (with back auditing), respectively, due to Hokamp and Pickhardt (2010). 'Without (with) back auditing' refers to the number of tax relevant periods subject to an audit, i.e. $\alpha = 0$ ($\alpha = 10$), in addition to the actual period. All figures are denoted in percent.

Table 3
Public goods provision changes.

Parameter	Tax collection ε_1	Auditing process ε_2	Fixed costs ε_3	Public good β	Voluntary MPCR $MPCR_{voluntary}$	Forced MPCR $MPCR_{forced}$	Maximum number of free-riders n_{max}
Period							
1	0.006	0.50	10,000	0.2000	0.198800	0.100000	5
5	0.006	0.25	10,000	0.2000	0.198800	0.150000	5
9	0.006	0.25	10,000	0.0010	0.000994	0.000750	1006
13	0.100	0.25	10,000	0.0010	0.000900	0.000750	1111
17	0.100	0.25	10,000	0.0050	0.004500	0.003750	222
21	0.100	0.25	0	0.0050	0.004500	0.003750	222
25	0.100	0.25	0	0.0075	0.006750	0.005625	148
29	0.100	0.25	0	0.0100	0.009000	0.007500	111
33	0.006	0.25	0	0.0100	0.009940	0.007500	100
37	0.006	0.25	0	0.2000	0.198800	0.150000	5
40	0.006	0.25	0	0.2000	0.198800	0.150000	5

Note: Four parameters are assigned to values (indicated by the gray-shaded area), that are, the effectiveness of 'tax collection', ε_1 (taken from Slemrod and Yitzhaki, 2002), the effectiveness of 'auditing process', ε_2, the 'fixed costs' of the tax authority, ε_3, and the effectiveness of 'public goods' sector, β. Given these figures 'marginal per capita returns' of voluntary ($MPCR_{voluntary}$) and forced ($MPCR_{forced}$) contributions to public goods as well as the maximum number of free-riders (n_{max}) are derived. 'Period' denotes tax relevant periods where changes take place, which is every fourth period. Changing parameters are denoted in bold. Fixed costs and the maximum number of free-riders are denoted in tokens and number of agents, respectively. Remaining figures are provided in percent.

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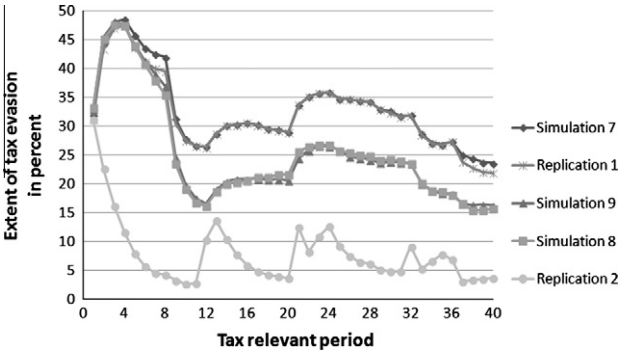


Fig. 2.1. Public goods provision and pareto-optimality ($\alpha = 0$).

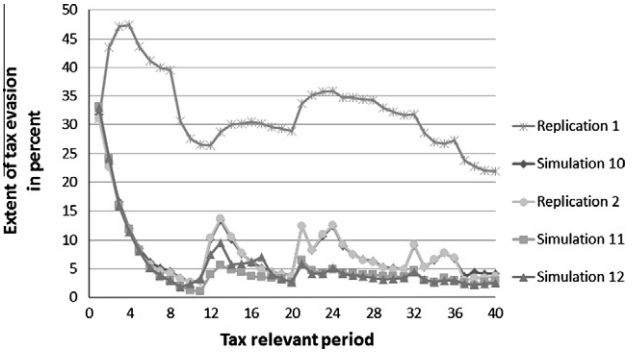


Fig. 2.2. Public goods provision and pareto-optimality ($\alpha = 10$). *Note*: Figure 2.1 provides without backauditing the extent of tax evasion under 'public goods provision' (Simulation 7), extended with 'age heterogeneity' and 'social norm updating' (Simulation 8), and, in addition, 'Pareto-optimality' (Simulation 9). Figure 4 shows with back auditing the extent of tax evasion under 'public goods provision' (Simulation 10), extended with 'age heterogeneity' and 'social norm updating' (Simulation 11), and, in addition, 'Pareto-optimality' (Simulation 12). Again, two benchmarks are provided, that are, Replications 1 and 2 reflecting reproductions of Scenarios 1 (without back auditing) and 2 (with back auditing), respectively, due to Hokamp and Pickhardt (2010). 'Without (with) back auditing' refers to the number of tax relevant periods subject to an audit, i.e. $\alpha = 0$ ($\alpha = 10$), in addition to the actual period. All figures are denoted in percent.

5. Conclusion

In this paper I have replicated and extended various features of agent-based tax evasion models with a view to analyze interaction dynamics that may result. In particular, I have implemented four extensions, (i) age heterogeneity, (ii) norm updating due to an age effect, (iii) provision of a pure public good and (iv) behavioral updating with respect to the Pareto-optimal provision of pure public goods. The dynamic effects of each extension have been analyzed *ceteris paribus* and jointly.

Findings include that age heterogeneity leads to fluctuations in income tax evasion, if and only if lapse of time effects (back auditing) or social norm updating are considered. Moral age effects, where older agents tend to be more compliant due to the evolution of social norms over their life cycle, may lead to less tax evasion, which is shown by a substantial vertical downward shift of income tax evasion dynamics. It is remarkable that these results confirm outcomes of Nordblom and Žamac (2012) and other scholars who analyze age effects and social interactions. Moreover, social norm updating may be linked to subject's audit experience (e.g. see Lipatov, 2012, with respect to social interaction dynamics and tax evasion); a feature that was employed to update agent's subjective audit probability. In addition, instead of fixed magnitudes tax administrative parameters may vary and depend on tax and penalty rates. However, these amendments rather delineate a future research agenda.

Next, findings with respect to interaction dynamics between the government as a provider of public goods and consumers, i.e. taxpayer agents, support the counterintuitive conclusion of Falkinger (1988, 1995) and Cowell (1992) that the provision of public goods leads to more tax evasion, provided that (voluntary) marginal per capita returns exceed a certain threshold. A precise estimation or calculation of turning points is left for future research. Another research task of interest may be an analysis of tax evasion dynamics due to individualizing the effectiveness of the public sector. Nonetheless, it is worth noting that the technical probability for a Pareto-optimal provision of public goods was rather small within the agent-based tax compliance simulation.

To my best knowledge, for the first time the Pareto-optimality concept is incorporated within an agent-based model and it is shown that, in fact, Pareto-optimal allocations of public goods do definitely emerge. Finally, back auditing and/or social norm updating not only dominate income tax compliance dynamics in any simulation scenario, but may also change findings

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that occur in scenarios in which back auditing and/or social norm updating is not applied. Therefore, future research on tax evasion dynamics should always incorporate lapse of time effects and social behavior patterns.

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Appendix A

The appendix provides a replication study and a mathematical appendix.

A.1. Replication study

To begin with, Table A1 provides a political cycle due to Hokamp and Pickhardt (2010) that is used to replicate their scenarios *ceteris paribus* employing another programming language.

In particular, Hokamp and Pickhardt (2010, Fig. 1, p. 548) and (cf., Fig. 2, p. 551) last line from above are reproduced.

Fig. A1 shows the extent of tax evasion according to (23) taking original data into account, denoted as 'Scenario 1' and 'Scenario 2', as wells as using replication data of prior scenarios, denoted as 'Replication 1' and 'Replication 2'. The

Table A1
Political cycle. *Source*: Hokamp and Pickhardt (2010, Table 1, p. 544).

Parameter	Tax rate θ	Penalty rate π	Tax complexity γ	Audit probability p
Period				
1	0.20	0.30	0.10	0.01
5	0.20	0.30	0.10	**0.03**
9	0.20	**0.45**	0.10	0.03
13	**0.30**	0.45	0.10	0.03
17	0.30	0.45	**0.20**	0.03
21	**0.40**	0.45	0.20	0.03
25	0.40	0.45	0.20	**0.04**
29	0.40	0.45	0.20	**0.05**
33	0.40	**0.50**	0.20	0.05
37	**0.30**	0.50	0.20	0.05
40	0.30	0.50	0.20	0.05

Note: 'Parameter' denotes' tax rate' θ, ' penalty rate' π, 'tax complexity' γ and 'audit probability' p. 'Period' denotes the tax relevant period when changes take place, which is every fourth period. Changing parameters are denoted in bold. All figures are denoted in percent.

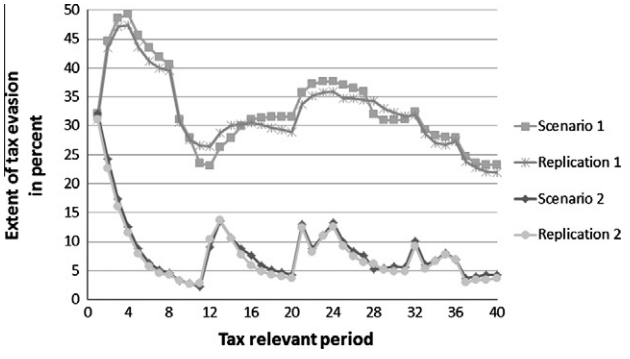


Fig. A1. Extent of tax evasion – Replication study. *Note*: Figure A1 shows the extent of tax evasion of a setting due to Hokamp and Pickhardt (2010) without and with lapse of time effects (Scenario 1 and Scenario 2, respectively) and their replications without and with lapse of time effects (Replication 1 and Replication 2, respectively). Lapse of time scenarios refer to ten tax relevant periods subject to an audit, i.e. $\alpha = 10$, in addition to the actual period. All figures are denoted in percent.

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replications confirm well-known income tax evasion dynamics, for instance, a substantial decrease of fiscal fraud at the end of 'Scenario 1' and 'Scenario 2' that is induced by lowering the tax rate. Results are in line with Andrei et al. (2011) and Pickhardt and Seibold (2011), who find strong evidence within their replication studies for lapse of time effects and an increasing extent of tax evasion at the beginning of 'Scenario 1', respectively. To highlight, Fig. A1 indicates support for the finding that back auditing dominates income tax evasion behavior.

A.2. Mathematical appendix

A.2.1. Comparative statics

Allingham and Sandmo (1972), without back auditing

$$NLT := \ln[\{(1-p)\cdot(1-MPCR_{voluntary})\cdot\theta\}/\{p\cdot(\pi\cdot(1-MPCR_{forced})-\theta\cdot(1-MPCR_{voluntary}))\}] \tag{A.1}$$

$$\frac{\partial x_i}{\partial W_i} = 1 > 0 \tag{A.2}$$

$$\frac{\partial x_i}{\partial \pi} = \frac{1}{\lambda_i\cdot\pi\cdot(1-MPCR_{forced})}\cdot\left(\frac{1-MPCR_{forced}}{\pi\cdot(1-MPCR_{forced})-\theta\cdot(1-MPCR_{voluntary})}+\frac{NLT}{\pi}\right) > 0 \tag{A.3}$$

$$\frac{\partial x_i}{\partial p} = \frac{1}{\lambda_i\cdot(1-MPCR_{forced})\cdot\pi\cdot p\cdot(1-p)} > 0 \quad (0<p<1) \tag{A.4}$$

$$\frac{\partial x_i}{\partial \theta} = -\frac{1}{\lambda_i\cdot\theta\cdot(\pi\cdot(1-MPCR_{forced})-\theta\cdot(1-MPCR_{voluntary}))} < 0 \tag{A.5}$$

$$\frac{\partial x_i}{\partial \lambda_i} = \frac{NLT}{\lambda_i^2\cdot(1-MPCR_{forced})\cdot\pi} > 0 \tag{A.6}$$

$$\frac{\partial x_i}{\partial \varepsilon_3} = 0 \tag{A.7}$$

Allingham and Sandmo (1972), with back auditing

$$\frac{\partial x_{i,t}}{\partial \pi} = -\left(\sum_{k=t-\alpha}^{t-1}\frac{\pi_k}{\pi^2}\cdot(W_{i,k}-x_{i,k})\right)+\frac{1}{\lambda_i\cdot\pi\cdot(1-MPCR_{forced})}\cdot\left(\frac{1-MPCR_{forced}}{\pi\cdot(1-MPCR_{forced})-\theta\cdot(1-MPCR_{voluntary})}+\frac{NLT}{\pi}\right)$$
$$=<>0 \tag{A.8}$$

Yitzhaki (1974), without back auditing, $F>1$

$$NLT_Y := \ln\left[\{(1-p)\cdot(1-MPCR_{voluntary})\}/\{p\cdot(F\cdot(1-MPCR_{forced})-(1-MPCR_{voluntary}))\}\right] \tag{A.9}$$

$$\frac{\partial x_i}{\partial \theta} = \frac{NLT_Y}{\lambda_i\cdot(1-MPCR_{forced})\cdot F\cdot\theta^2} > 0 \tag{A.10}$$

$$\frac{\partial x_i}{\partial F} = \frac{1}{\lambda_i\cdot F\cdot\theta\cdot(1-MPCR_{forced})}\cdot\left(\frac{1-MPCR_{forced}}{F\cdot(1-MPCR_{forced})-(1-MPCR_{voluntary})}+\frac{NLT_Y}{F}\right) > 0 \tag{A.11}$$

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